

AI Engineer Assessment — Customer Support Chatbot

The Situation

You're an AI engineer at **Meridian Electronics**, a mid-size company that sells computer products — monitors, keyboards, printers, networking gear, and accessories. The support team currently handles all customer inquiries by phone and email, and leadership wants to explore whether an AI-powered chatbot could handle common requests: checking product availability, helping customers place orders, looking up order history, and authenticating returning customers.

The backend team has already built an internal services layer and exposed it as an **MCP server** so that an LLM-based chatbot can call into existing business systems without direct database access:

None

```
MCP_SERVER_URL=https://order-mcp-74afyau24q-uc.a.run.app/mcp
Transport: Streamable HTTP
```

Your engineering manager has asked you to **build a working prototype** and present it to the team by the end of day. The VP of Customer Experience will also be watching — she wants to understand whether this is worth investing in before next quarter's planning.

Your Assignment

Build a deployable, production-ready prototype of the customer support chatbot.

"Production-ready" means: if the demo goes well, engineering should be able to take your code, do a security review, and push it to production without a rewrite. Think clean architecture, appropriate error handling, tested behavior, and thoughtful decisions — not just "does it work."

Constraints

- **LLM:** Use a cost-effective model (flash or mini tier — e.g., Gemini Flash, GPT-4o-mini, Claude Haiku). The business case only works if per-conversation costs stay low.

- **UI:** At minimum, a functional chat interface (Gradio, Streamlit, or equivalent). If you have time, a full UI framework (Next.js, etc.) would strengthen the demo.
- **Deployment:** At minimum, deploy to HuggingFace Spaces so the team can try it live. Bonus for deployment to Vercel, GCP, AWS, or Azure.
- **Scope:** Connect to the MCP server, discover its available tools, and build a chatbot that uses them to serve Meridian's customers. The MCP tools define what the chatbot can do — explore them and make sure your bot handles those workflows well.

Time

You have **3 hours**. You won't be able to do everything perfectly — that's by design. How you prioritize is part of what we're evaluating. Budget your time accordingly.

Deliverables

Video 1 — Kickoff (submit by: **{START TIME + 30 MINS}**)

You've just read the brief. Before writing any code, record a short video:

- What's the business problem you're solving for Meridian?
- What's your plan of attack? What will you build first, and what might you cut if you run out of time?

Video 2 — Mid-point check-in (submit by: **{START TIME + 1 HR 45 MINS}**)

You're partway through. Record a video covering:

- What you've built so far and the decisions you've made
- What you've done to correct or improve AI-generated code
- Any challenges you've hit and how you're working through them

Video 3 — Final Presentation (submit by: **{START TIME + 3 HRS}**)

You're presenting to **your engineering manager and the VP of Customer Experience** to pitch whether this prototype should move forward. This is the most important deliverable.

Length: 3–10 minutes

Your presentation should include:

- A live demo of the chatbot handling real customer scenarios
- A walkthrough of the code and technical architecture — focus on important decisions and show that you understand what the code is doing (not just that it runs)
- Honest assessment: what works well, what are the limitations, and what would you improve with more time?

Final submission

Turn in:

- **Video 3**
- **GitHub repo**
- **Deployed chatbot URL**
- **Screenshots of the solution in action**

Timing

- Submissions are due at the posted deadline
- Full credit for on-time submission
- Credit is reduced for every 20 minutes late